Experiment 5

Report Generation Using Relational and Multidimensional Data Using Microsoft Power BI

AIM:

To generate relational and multidimensional reports using Microsoft Power BI by utilizing transactional and dimensional data models, enabling effective business analysis, reporting, and decision-making.

OBJECTIVES:

- To understand the concepts of relational and multidimensional reporting in Business Intelligence.
- To generate relational reports using transactional business data.
- To create multidimensional reports using hierarchies and OLAP concepts.
- To apply sorting, filtering, grouping, and aggregation to business data.
- To perform drill-down and drill-up operations for detailed analysis.
- To create comparative visualizations for analyzing sales, profit, and customer performance.
- To design interactive dashboards using KPI cards and slicers.
- To interpret business insights from reports and visualizations.

TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

DATASET USED: Global Superstore Dataset.xlsx. https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

THEORY:

Business Intelligence (BI) reporting is the process of transforming raw business data into meaningful information for analysis and decision-making. Microsoft Power BI is a Business Intelligence tool that enables users to connect, transform, analyze, and visualize data through interactive reports and dashboards.

Relational Reporting: Relational reports display detailed transactional records in a tabular format. They are useful for examining individual orders, customers, products, sales, and profits. Power BI provides table visuals, sorting, and filtering features to generate relational reports.

Multidimensional Reporting: Multidimensional reports allow data to be analyzed across multiple dimensions, such as Region, State, City, Product, Customer, and Time. Matrix visuals and hierarchies help users summarize and explore business performance at different levels.

OLAP (Online Analytical Processing): OLAP enables users to analyze data from different perspectives. Drill-down provides more detailed information, while drill-up returns to a higher level of summarized information.

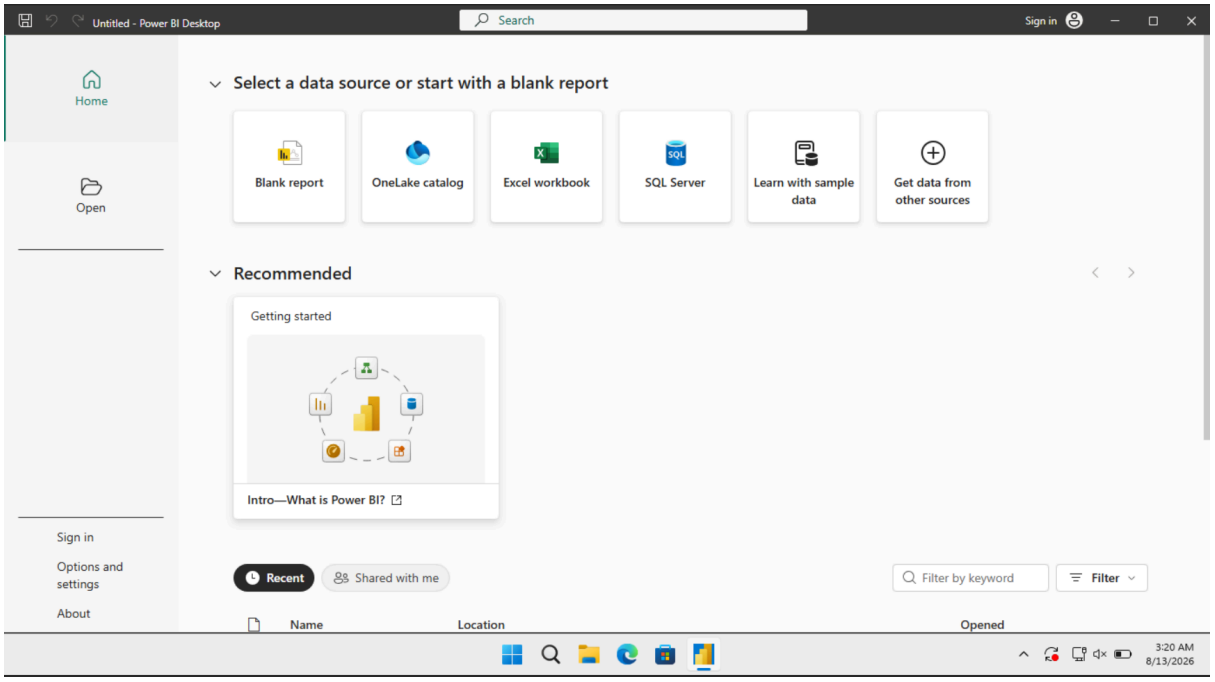
Data Visualization: Power BI supports bar charts, column charts, line charts, tables, matrices, and KPI cards to represent business data effectively.

Interactive Dashboard: An interactive dashboard combines multiple visualizations and slicers into a single report page. Slicers allow users to filter information by dimensions such as Region, Category, and Year.

PROCEDURE:

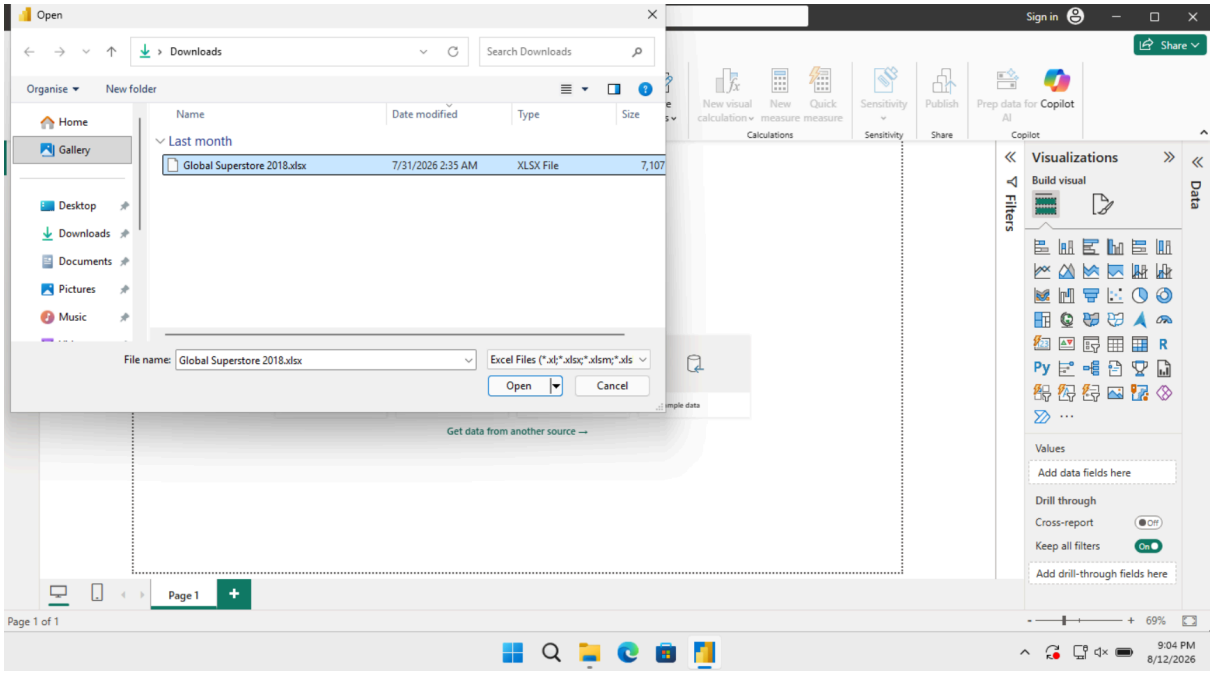
Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.



Step 2: Import the Global Superstore dataset.

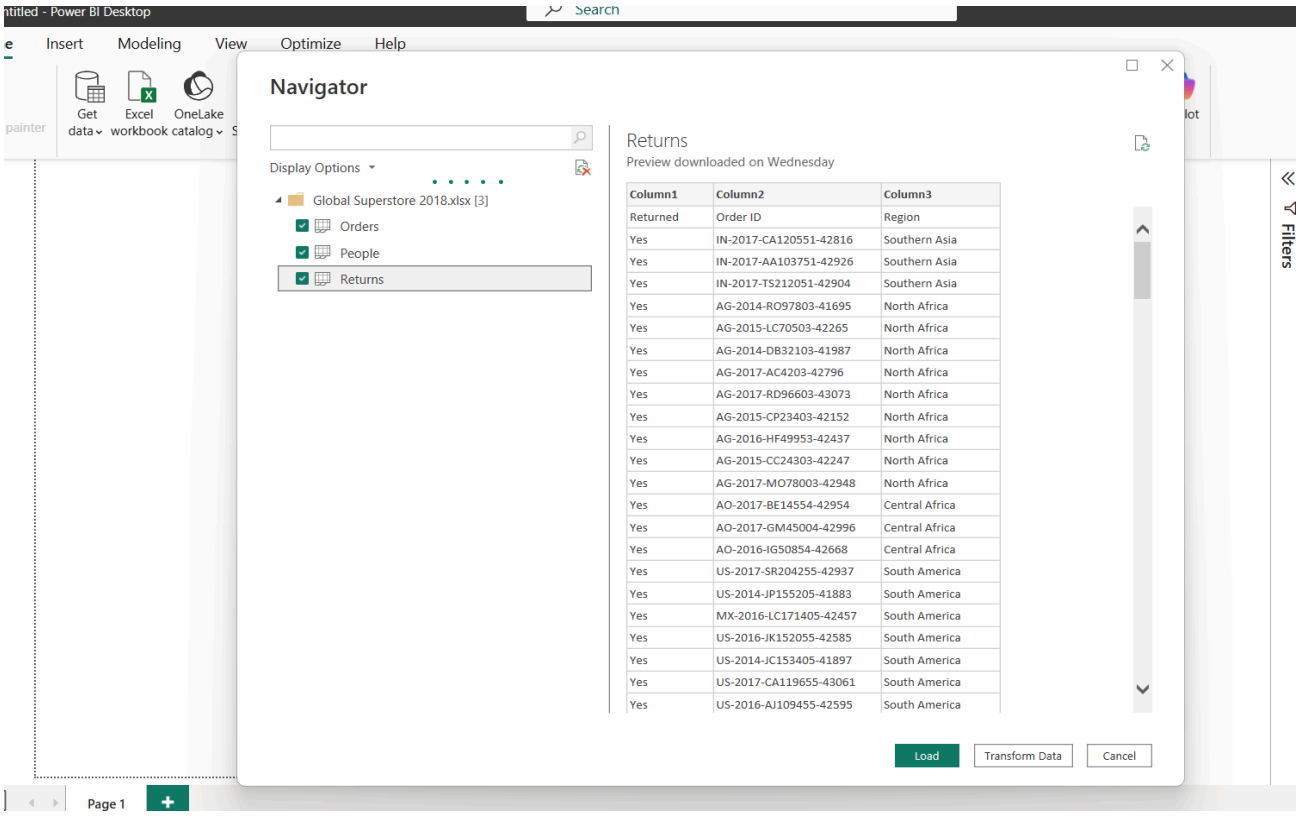
Home > Get Data > Excel Workbook Select:



Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:



• Orders

• Returns

• People

Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables

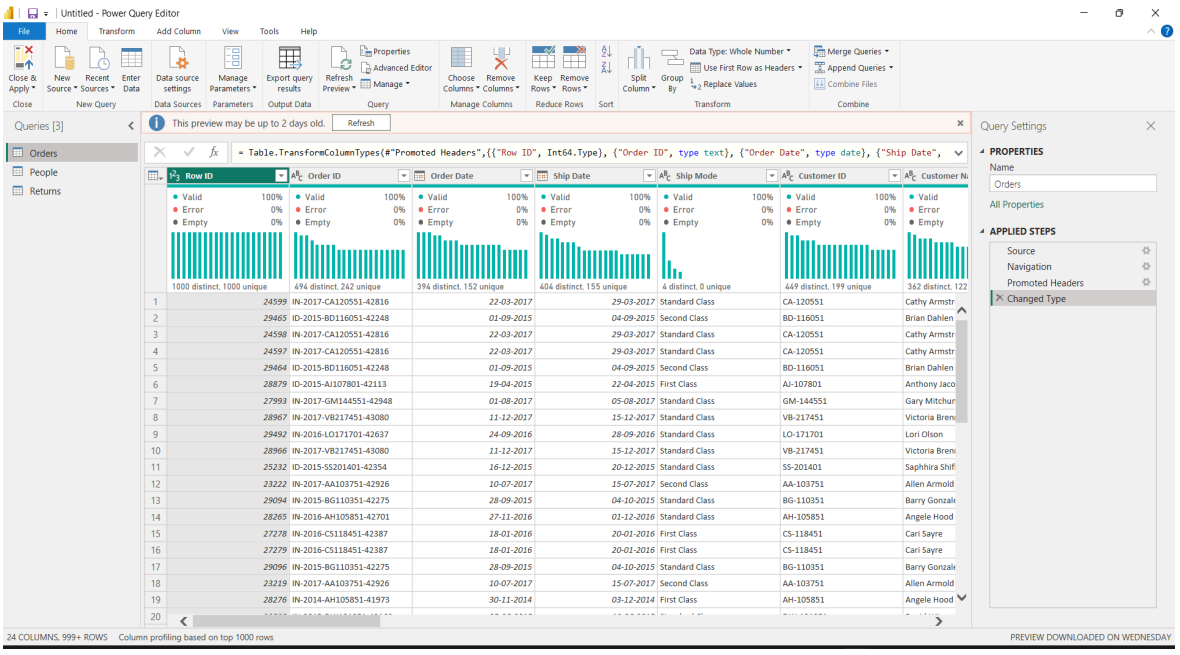
Step 4: Analyse the Orders table.

Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity
- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.



Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data. Organize the attributes into:

A. Fact Table: Fact_Sales

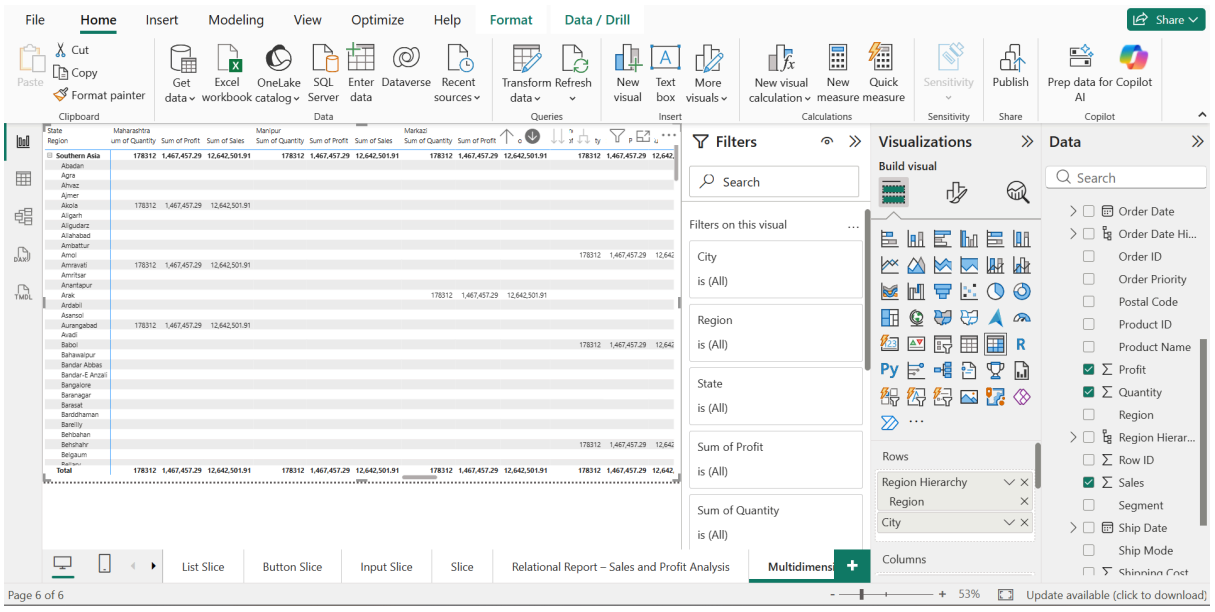
B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

This classification forms the basis of the Star Schema.

Part C: Generate Multidimensional Report

Step 7: Create a Hierarchy

In the Dim_Region table, create a hierarchy named Region Hierarchy using the following order:



Region → State → City

Step 8: Insert a Matrix Visual

Select the Matrix visual from the Visualizations pane and place it on the report canvas.

Step 9: Add the Hierarchy and Measures

Add Region Hierarchy to the Rows section of the Matrix visual.

Add the following measures or numeric fields to the Values section:

- Sales
- Profit
- Quantity

Ensure that Sales, Profit, and Quantity are aggregated appropriately, usually using Sum.

Step 10: Perform Drill-down and Drill-up

Use the hierarchy controls to navigate from Region to State and then City. Use Drill-up to return to higher levels of aggregation.

Step 11: Analyze Aggregated Performance

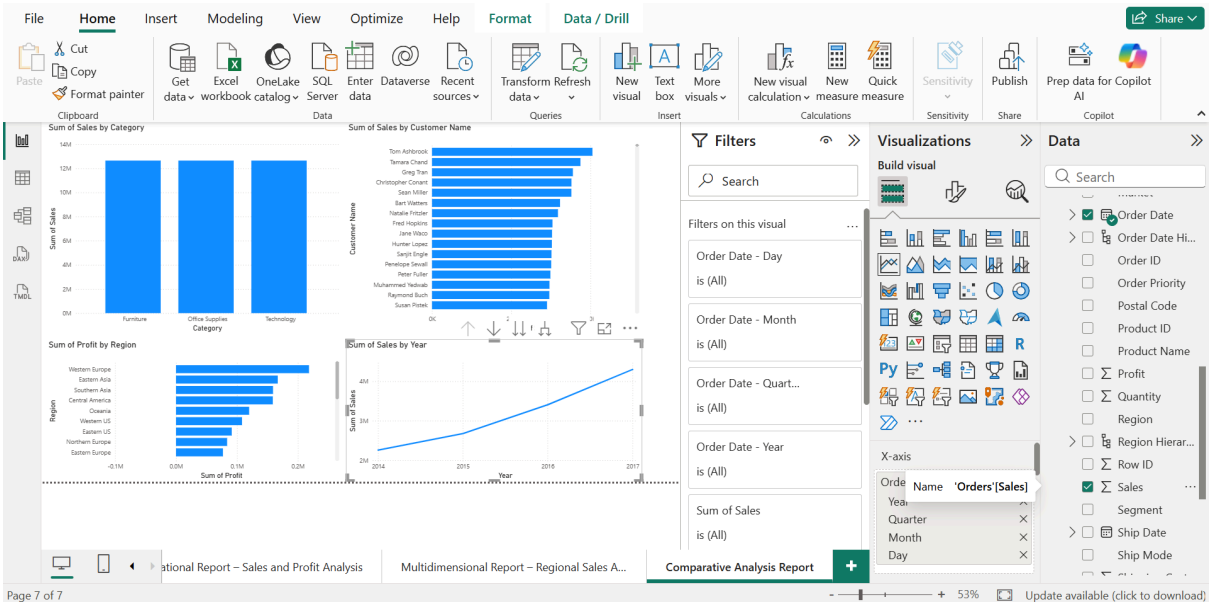
Compare sales, profit, and quantity across different geographical levels to identify regional differences in business performance.

Part C: Generate Comparative Analysis Report

Step 12: Create Sales by Category Chart

Insert a Clustered Column Chart. Add Category to the X-axis and Sales to the Y-axis. Set the aggregation of Sales to Sum.

Set the title to Sales by Category.



Step 13: Create Profit by Region Chart

Insert a Clustered Bar Chart. Add Region to the Y-axis and Profit to the X-axis. Set Profit aggregation to Sum. Set the title to Profit by Region.

Step 14: Create Top 10 Customers by Sales Chart

Insert a Clustered Bar Chart. Add Customer Name to the Y-axis and Sales to the X-axis.

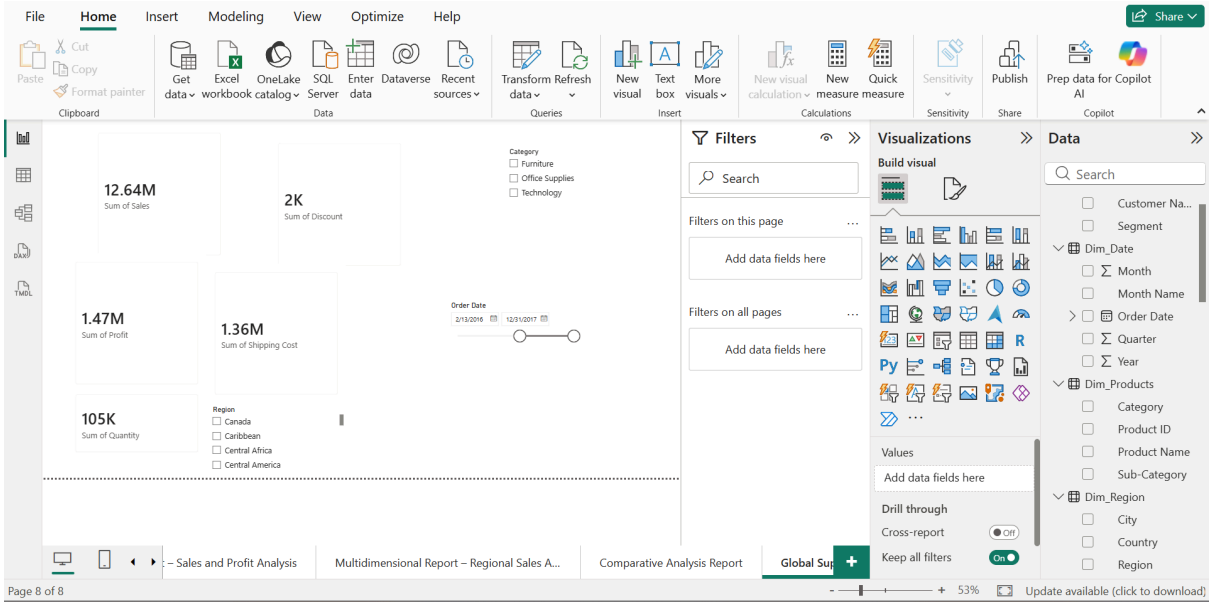
Apply a Top N filter with the following settings:

- Filter type: Top N
- Show items: Top 10
- By value: Sales
- Aggregation: Sum of SalesApply the filter and sort the chart by Sales in descending order.
- Set the title to Top 10 Customers by Sales.
- Step 16: Create Year-wise Sales Trend
- Insert a Line Chart. Add Order Date to the X-axis and Sales to the Y-axis.
- Select the Year level of the Date hierarchy and aggregate Sales using Sum.
- Set the title to Year-wise Sales Trend

Step 15: Compare the Visualizations

Compare the results of the four visualizations to understand sales distribution by category, profitability by region, customer contribution, and yearly sales trends.

Part D: Create an Interactive Dashboard



Step 16: Create Total Sales KPI Card

Insert a Card visual and add Sales with Sum aggregation.

Set the title to Total Sales.

Step 17: Create Total Profit KPI Card

Insert another Card visual and add Profit with Sum aggregation.

Set the title to Total Profit.

Step 18: Create Total Orders KPI Card

Insert another Card visual and add Order ID. Set the aggregation to Distinct Count to count unique orders rather than individual product rows. Set the title to Total Orders.

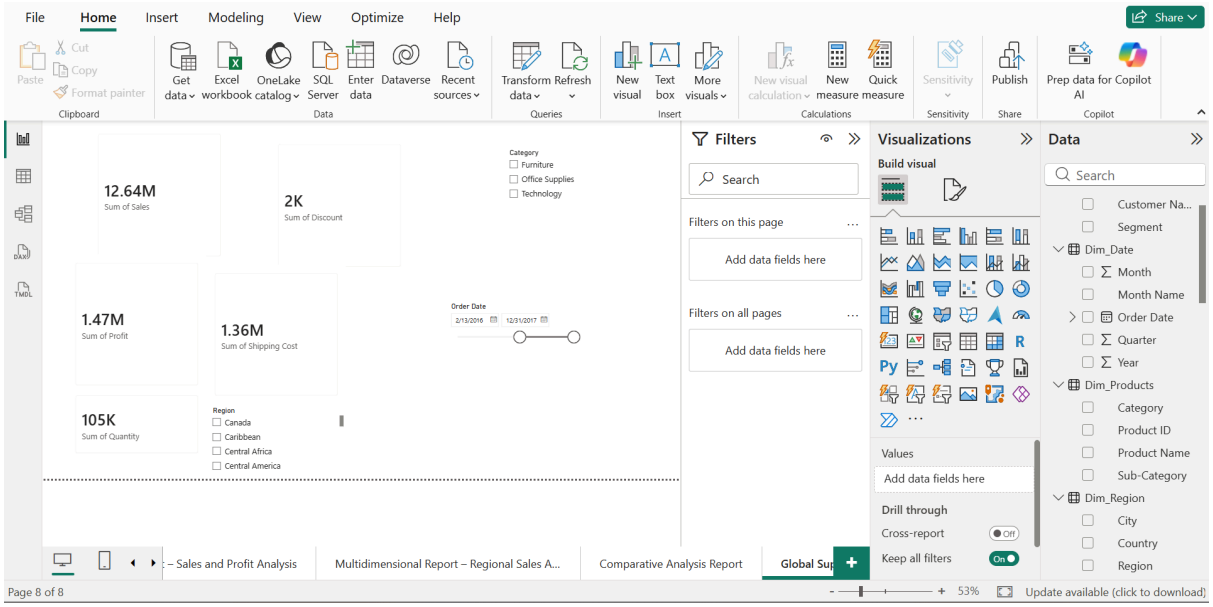
Step 19: Add Region Slicer

Insert a Slicer visual and add Region from the appropriate region dimension table.

Set the title to Filter by Region.Step 22: Add Category Slicer

Insert another Slicer visual and add Category from the appropriate product or sales table.

Set the title to Filter by Category.



Step 20: Add Year Slicer

Insert another Slicer visual and add Order Date. If using the Date hierarchy, select the Year level.

Set the title to Filter by Year.

Step 21: Arrange the Dashboard

Arrange the three KPI cards, slicers, and four charts into a clear and organized dashboard layout.

Set the dashboard title to:

Global Superstore – Business Performance Dashboard

Step 22: Verify Interactive Functionality

Select different values in the Region, Category, and Year slicers. Verify that the relevant charts and KPI cards update according to the selected filters.

If any visual does not respond as expected, check the visual interactions and data relationships.

Step 23: Save the Report

Save the completed Power BI project using Ctrl + S. Keep the report file in PBIX format for future editing and Presentation.

OBSERVATIONS

1. A relational report was generated using a table visual to display transactional details such as Order ID, Customer Name, Product Name, Region, Sales, and Profit.
2. Sorting and filtering helped organize records and focus on selected regions.
3. The report data was successfully exported for further analysis.
4. A multidimensional report was generated using a Matrix visual and a Region → State → City hierarchy.
5. Drill-down and drill-up operations enabled analysis at different geographical levels.
6. The Sales by Category chart helped compare sales across product categories.

7. The Profit by Region chart enabled comparison of regional profitability.
8. The Top 10 Customers by Sales chart highlighted customers contributing the highest sales.
9. The Year-wise Sales Trend chart represented changes in sales over time.
10. KPI cards displayed Total Sales, Total Profit, and Total Orders.
11. Region, Category, and Year slicers provided interactive filtering functionality.
12. The dashboard enabled consolidated analysis of business performance.

RESULT:

OLAP operations were successfully performed on the Global Superstore dataset using Microsoft Power BI. Slice, Dice, Drill-down, and Drill-up operations were applied to analyze sales and profit across different dimensions, providing useful insights for business analysis.

LEARNING OUTCOMES:

- Understood the concept and purpose of OLAP in Business Intelligence.
- Learned how multidimensional data can be analyzed from different perspectives.
- Gained practical experience with Slice and Dice operations.
- Learned how to navigate between summarized and detailed information using Drill-down and Drill-up.
- Learned how to use Power BI filters and hierarchies for analytical exploration.
- Understood how sales and profit can be compared across different dimensions.
- Gained experience in creating interactive OLAP-based reports.
- Learned how OLAP analysis can support business decision-making.